

RAMCO INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
ACADEMIC YEAR 2017-18
ME6811-PROJECT WORK

Sl.No.	Project title	PO's relevance	PSO's relevance	Outcome
New product development				
1.	Analysis and Testing of a Forced Convection Solar Dryer for Tropics	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,3,4	Presentation at National Conference on Recent Advancement in Mechanical Engineering'18", Kamaraj College of Engineering and Technology, Virudhunagar
2.	Humanoid Flexo Grab	POs – 1,2,3,4,5,8,9,10,11,12	PSOs – 1,2	Presentation at National Conference on Research Advances in Mechanical Engineering, P.S.R. Engineering College, Virudhunagar.
3.	Design, Fabrication and Analysis of 3-Axis CNC Milling Machine	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,4	Presentation at National Conference on Research Advances in Mechanical Engineering, P.S.R. Engineering College, Virudhunagar.
4.	Design, fabrication and Analysis of Gearless Transmission with Scotch Yoke Mechanism	POs – 1,2,3,4,5,9,10,11,12	PSOs -1,2,4	Presentation at TNSCST Sponsored National Conference on Research Advances in Mechanical Engineering.
5.	Design and Analysis of Roll Cage for All Terrain Vehicle.	POs – 1,2,3,4,5,9,10,11,12	PSOs -1,2,4	• Presented this project in SAE-BAJA Virtual round held at Chitkara University, Punjab. • Participated in the Mega ATV Championship held at Nashik, Maharashtra.
6.	Design and Analysis of Transmission System and Knuckle for All Terrain Vehicle	POs – 1,2,3,4,5,9,10,11,12	PSOs -1,2,4	• Presented this project in SAE-BAJA Virtual round held at Chitkara University, Punjab. • Participated in the Mega ATV Championship held at Nashik, Maharashtra.

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7.	Fabrication and Experimental Analysis of various Shape of Shell and Tube Heat Exchanger	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,3	Presentation at National Conference on Research Advances in Mechanical Engineering, P.S.R. Engineering College, Virudhunagar.
8.	Design, Fabrication and Analysis of Abrasive Jet Machine	POs – 1,2,3,4,5,9,10,11,12	PSOs -1,2,4	Presentation at National Conference on Research Advances in Mechanical Engineering, P.S.R. Engineering College, Virudhunagar.
9.	Design and Development of Autonomous Robot for Heterogeneous Application	POs – 1,2,3,4,5,8,9,10,11,12	PSOs – 1,2,4	• Presentation at National Conference on Research Advances in Mechanical Engineering, P.S.R Engineering College, Sivakasi. • Project carried out at PACR Polytechnic college
10.	Performance Characteristics of Pyramid Shaped Solar Still	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,3,4	A comparative study on conventional and pyramid shaped solar still efficiency was studied
Projects in Identified researches				
11.	Influence of Ceramic and Metallic Reinforcement on Mechanical Properties of Stir Casting Aluminium Matrix Composite	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at “International Conference on Engineering and Advancement in Technology’2018” , Sri Krishna College of Technology, Coimbatore
12.	Aggreivation of Mechanical and Tribological Properties of Aluminium 6063 by Severe Plastic Deformation	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at National Conference on Recent Advancement in Mechanical Engineering’18”, Kamaraj College of Engineering and Technology, Virudhunagar.

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13.	Effect of Boron Carbide on Enhancing the Mechanical Properties of Copper Nano Composites	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at International conference on Recent Innovation in Mechanical Engineering (ICRIME 2K19) conducted by Francis Xavier Engineering College, Tirunelveli.
Projects in specific applications				
14.	Reduction of Fuel Consumption and Pollution in Automobile using HHO	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,3	Presentation at National Conference on Research Advances in Mechanical Engineering, P.S.R Engineering College, Sivakasi
15.	Smart Helmet using Arudino for Two – Wheeler Riders.	POs – 1,2,3,4,5,8,9,10,11,12	PSOs – 1,2	Presentation at “National Conference in Innovation in Communication Technology for Social Development”, Ramco Institute of Technology, Rajapalayam.
16.	Design and Experimental Analysis of Direct Evaporative Cooling System for Different Cooling Pad Materials.	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,3,4	Presentation at 5 th National Conference on “Research Advances in Mechanical Engineering”, P.S.R Engineering College, Sivakasi
17.	Distillation Of Waste Water By Using Biogas Extracted From Kitchen Waste	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,3	The process of Bio gas extraction from kitchen waste and water distillation is studied and performed.
Review/Experimental studies in emerging mechanical sciences				
18.	Finite Element Modelling and Experimental Investigation of GFRP Aircraft Wing Structure under Static Loading Condition	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,4	Presentation at National Conference on Recent Advancements in Mechanical Engineering at Kamaraj College of Engineering and Technology ,Virudhunagar.

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19.	Performance Studies of Single Basin Solar Still by varying the Concentration of Zinc Oxide Nano Particle.	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,3,4	• Presentation at “VIII International Conference on Engineering and Advancement in Technology” - 2018, Sri Krishna College of Technology, Coimbatore • Received I prize in ICT Academy
20.	Analysis of Heat Transfer Rate using Nano Fluids in Heat Exchanger.	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,3,4	Presentation at International Conference on Design, Analysis, Manufacturing and Simulation, Saveetha School of Engineering,, Chennai
21.	Performance Enhancement of Solar Parabolic Concentrator using Phase Change Material	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,3,4	Presentation at 5 th National Conference on “Research Advances in Mechanical Engineering”, P.S.R Engineering College, Sivakasi
22.	Experimental Studies on Formability of Light Weight Alloy AA6082 by Single Point Incremental Forming Process	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	• Presentation at National Conference on Research Advances in Mechanical Engineering, P.S.R. Engineering College, Virudhunagar. • Presentation at AIMTDR2018 International conference
23.	Machining Studies on Aluminium Metal Matrix Composites Using Optimization Techniques	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at National Conference on Research Advances in Mechanical Engineering, P.S.R. Engineering College, Virudhunagar.
24.	Experimental Studies on Mechanical and Wear Characteristics of Powder Metallurgy Processed Aluminium Composites	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Characteristics of Powder metallurgy processed Aluminium composites are experimentally studied.

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25.	Fabrication and Experimental Analysis of Fibre Glass Reinforced Polymer (FRP) on Railway Component	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2,4	Railway berth is fabricated using FRP and properties experimentally verified
26.	Machining Studies using Aluminium Alloy in Milling	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Process Parameters and characteristics of Aluminium Alloy in Milling is studied
27.	Experimental Studies on Tribological properties of ECAPed Aluminium Metal Matrix Composite	POs – 1,2,3,4,5,9,10,11,12	PSOs – 1,2	Presentation at National Conference on Recent Advancements in Mechanical Engineering at Kamaraj College of Engineering and Technology, Virudhunagar
Solution to Industrial problems based Projects				
28.	Design, Fabrication and Analysis of Portable Crane for a Construction Company.	POs - 1,2,3,4,5,6,7,9,10,11,12	PSOs-1,2,4	• M/S Madhura Builders & Architectures, Sankarankoil • Patent Filled Application number: 201841016885
29.	Energy Auditing on Compressor systems in Cement Industry for Energy Conservation	POs -1,2,4,5,8,9,10,11,12	PSOs-2,4	• "Energy Auditing on Compressor Systems in Cement Industry for Energy Conservation" • National Conference on VLSI, Optical Communication, Image Processing, Communication and Embedded Systems, Kalasalingam University, Krishnankoil, Tamilnadu.
30.	Design, Fabrication and Analysis of Rocket Cracker Nozzle Chocking Machine for Fireworks Industries	POs -1,2,3,4,5,6,7,9,10,12	PSOs-1,2,4	• Vinayagar Fireworks Factory, Sivakasi • “National Conference in Innovation in Communication Technology for

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				Social Development", Ramco Institute of Technology, Rajapalayam.
31.	E-Motorcycle Conversion and Autonomous Braking System	POs -1,2,3,4,5,6,7,9,10,11, 12	PSOs-1,2	• Sri Ram Motors Pvt.Ltd, Sattur. - International Conference on Trends in Engineering' • PSN college of Engineering and Technology, Tirunelveli.
32.	Fabrication of Labour Moving Vehicle and design of Hydraulic Half ale Pressing machine (Semi-Automatic).	POs -1,2,3,4,5,6,7,9,10,12	PSOs-1,2	Sri Vishnu Sankar Mill Limited, Rajapalayam
33.	Cotton Fibre Dust Collector.	POs -1,2,3,4,5,6,7,9,10,12	PSOs-1,2	Ramco Mills,Rajapalayam
34.	Pneumatic Automated Material Handling System to Lift and Place of Cup and Dhoop Pieces	POs -1,2,3,4,5,6,7,9,10,12	PSOs-1,2,4	VST Engineering Solutions Madurai - 625008
35.	Design and Analysis of Foreign Particle Removal Technique in Belt Conveyor	POs -1,2,3,4,5,6,7,9,10,12	PSOs-1,2	RAMCO Cements, R.R.Nagar, Virudhunagar
36.	Energy Audit in Various Mechanical Components.	POs -1,2,4,5,8,9,10,11,12	PSOs-2,4	ADR Polymers, Shed No 150, SIDCO Industrial Estate, Kappalore, Madurai.
37.	Design and Fabrication of Pedal Operated Spin Patrol Vehicle	POs - 1,2,3,4,5,6,7,9,10,11,12	PSOs-1,2	YES, Sandhya Spinning Mill Limited, Rajapalayam
38.	Fixture Fabrication of Vertical Chamfering Machine	POs -1,2,3,4,5,6,7,9,10,11, 12	PSOs-1,2	Bharat Heavy Electricals Limited (BHEL) Trichy